

Managing Secrets via Amber

Maintaining a secret in the open source world is quite tough. Amber provides a simple and authenticated solution for protecting data as 'secrets' while also enabling access to it, as and when needed. This article gives a demo of this, by using Git along with Amber.



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Amber is a secret management solution based on public key cryptography. Systems that use Amber allow any member in a team to add or update secrets. Only the person who has the private or secret key has the ability to decrypt this secret, making it pretty useful in the following scenarios:

- Usage in continuous integration (CI) systems like GitHub actions, GitLab, Bitbucket pipelines, etc.
- Development teams that comprise multiple sub-teams where individual members might have to add secrets without getting hold of the central secret key.

There are various alternatives with different features in the open source world that solve the same problem. Some of

these are: SOPs, credstash, sealed-secrets, Vault and cryptic.

What makes Amber different from them is the simplicity it offers. It neither has complex configuration, nor does it depend on any external dependencies like GPG. The entire program is available as a single binary code and uses sodiumoxide for the public key encryption.

When a program like Git is used with Amber, the entire history of when a secret was added, modified and used can be known.

Installation

The executables can be downloaded from the GitHub repository page: <https://github.com/fpco/amber>